Laryngeal elevation in native and non-native lexical tone production

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1 Motivation

Acoustic studies show native and non-native speakers differ in tone precision, but **articulation differences** remain understudied.

Both laryngeal elevation (extrinsic) and cricothyroid contraction (intrinsic) contribute to f0 modulation.

Laryngeal ultrasound offers a novel window into extrinsic laryngeal control during tone production [1,2,3].

Hypothesis: Learning to produce tones involves reweighting articulatory strategies, shifting from extrinsic toward intrinsic f0 control.

Prediction: Elevation–f0 correlations are weaker in native than non-native speakers during tone production.

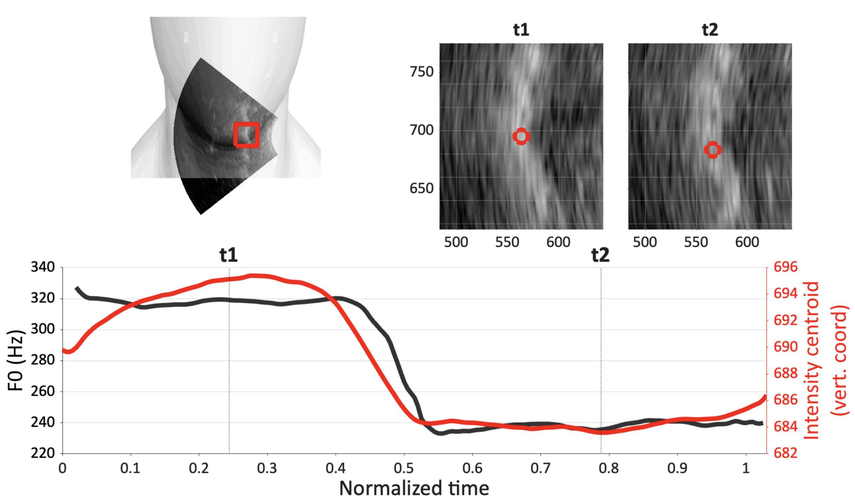
2 Coronal-orientation laryngeal ultrasound

Four participants: 2 native Mandarin speakers, 2 phonetically trained non-native speakers.

Tasks: high-to-low (HL) and low-to-high (LH) pitch sweeps + four Mandarin tones on *ma*.

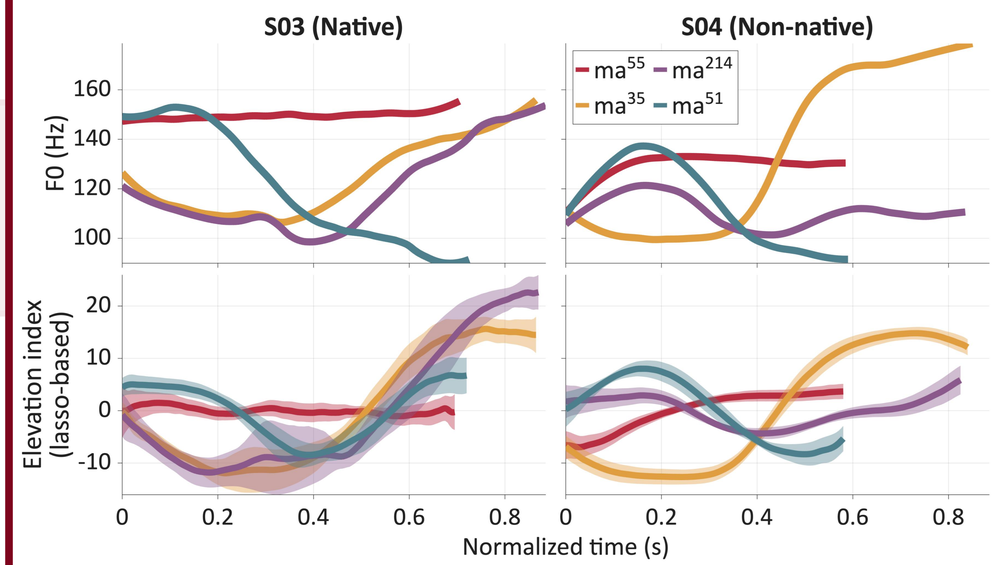
Two measures of laryngeal elevation

- *Centroid*: tracks vertical center of pixel intensity
- *Lasso-weighted sum*: weighted sum of pixel intensity sampled on a semipolar image grid, with weights from averages of lasso regressions predicting f0.

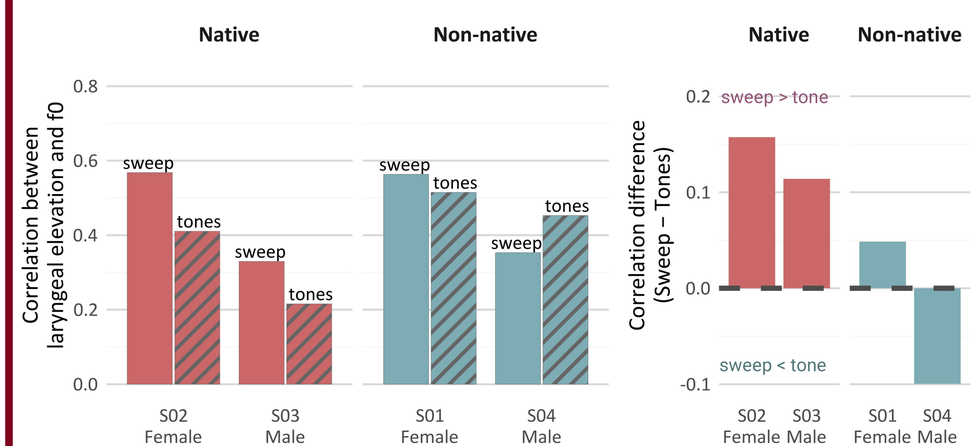


3 Results: native and non-native speakers differ in how laryngeal elevation relates to f0 across tasks.

1. Laryngeal elevation positively correlates with f0 across participants and tasks (replicates prior work [1, 4])



2. Native speakers show **stronger elevation–f0 correlations in sweeps than tones**; non-native speakers show **weaker or reversed** correlations.



4 Discussion

Assumption: Pitch sweeps reflect maximal utilization of laryngeal elevation, providing a baseline reference for extrinsic laryngeal control.

In sweeps: Strong positive correlation between laryngeal elevation and f0 → extrinsic control is active and available in both groups.

In tones: Such correlation drops substantially for native but not non-native speakers for tones.

Interpretation: Native speakers have learned to exploit intrinsic (cricothyroid) control of f0 as well, and may prefer it in normal speech, consistent with least-effort and/or motor equivalence principles.

5 Take-aways

① **Coronal-orientation laryngeal ultrasound** can track laryngeal elevation, which correlates with f0.

② **Native and non-native speakers differ** in elevation–f0 coupling across tasks.

③ **Speech learning may involve a reweighting of articulatory strategies.**

6 References

[1] Moisik, S. R., Lin, H., & Esling, J. H. (2014). A study of laryngeal gestures in Mandarin citation tones using simultaneous laryngoscopy and laryngeal ultrasound (SLLUS).
[2] Brunelle, M., Schweizer, D., & Ahn, S. (2025). An ultrasound study of the role of larynx height and tongue movement in the production of Javanese registers.
[3] Schweizer, D., Brunelle, M., Ahn, S., & Audet, A. (2025). A laryngeal and lingual ultrasound study of the Canadian French voicing contrast.
[4] Honda, K., Hirai, H., Masaki, S., & Shimada, Y. (1999). Role of Vertical Larynx Movement and Cervical Lordosis in F0 Control.

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